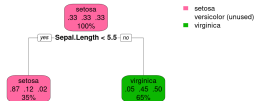
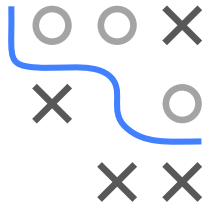


Introduction to Machine Learning

CART

Growing a Tree

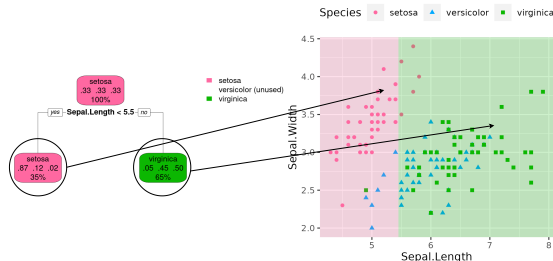
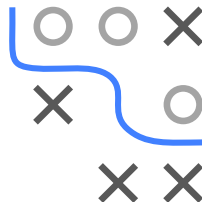


Learning goals

- Understand how a tree is grown by an exhaustive search
- Know where and how the split point is set

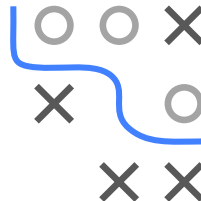
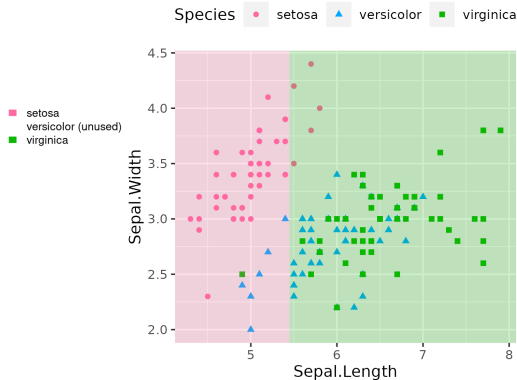
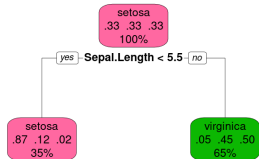
TREE GROWING I

- We start with an empty tree, a root node that contains all the data. Trees are then grown by recursively applying **greedy** optimization to each node $\mathcal{N}$.
- Greedy means we do an **exhaustive search**: Ideally, all possible splits of $\mathcal{N}$ on all possible points t for all features x_j are compared in terms of their empirical risk $\mathcal{R}(\mathcal{N}, j, t)$.
- The training data is then distributed to child nodes according to the optimal split and the procedure is repeated in the child nodes.



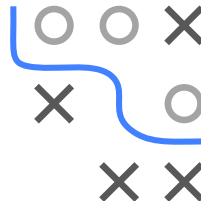
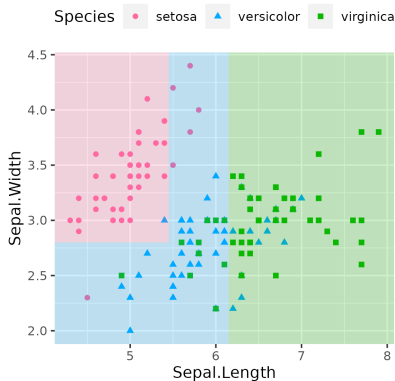
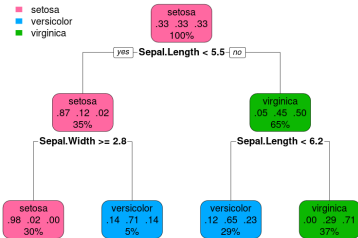
TREE GROWING I

- 1 Start with a root node of all data.
- 2 Search for feature and split point that minimizes the empirical risk in child nodes – makes label distribution more homogenous.



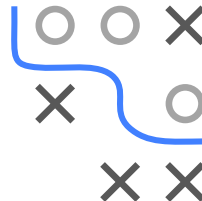
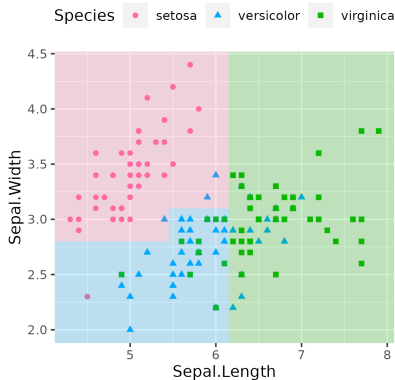
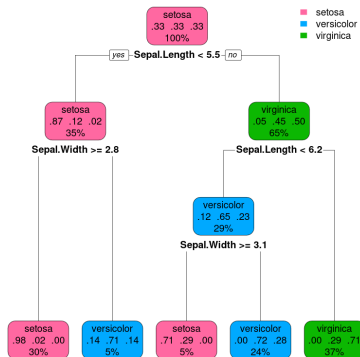
TREE GROWING I

- 3 Proceed recursively for each child node: Select best split and divide data from parent node into left and right child nodes.

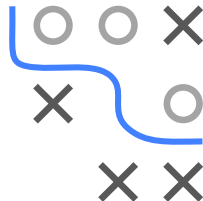
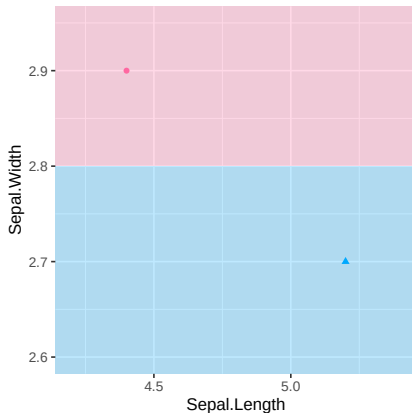


TREE GROWING I

- 4 Repeat until we reach a stop criterion, e.g., until each leaf cannot be split further.



SPLIT PLACEMENT I

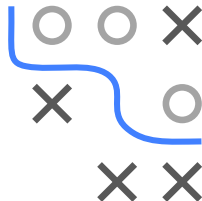
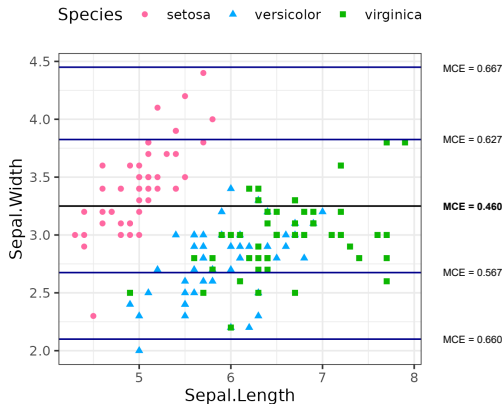


Splits are usually placed at the mid-point of the observations they split:
the large margin to the next closest observations makes better
generalization on new, unseen data more likely.

FINDING THE SPLIT I

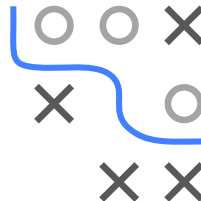
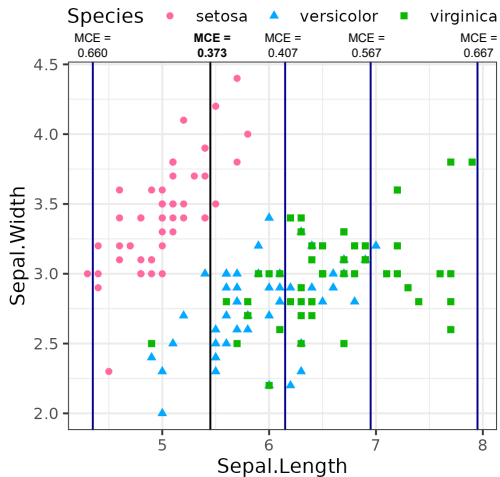
Assume we split the data so that the misclassification error (MCE) is minimal through the splitting.

First, we check a set of potential splits for `Sepal.Width`



FINDING THE SPLIT I

Then we check a set of potential splits for Sepal.Length



FINDING THE SPLIT I

- We take the split with lowest MCE: $\text{Sepal.Length} = 5.5$
- In real life, we actually search over many more splitting points. Common strategies involve: a) Searching over all possible split points (exhaustive search), b) searching quantile-wise
- MCE is rarely used, we will cover split criteria in detail later.

